

Sudhir Varanasi

Consultant
Software Methods & Data Analytics

48, Alonso Close, Chellaston, Derby, DE73 5AX •

(+44) 7466296348 •

Sudhir.varanasi@gmail.com •

linkedin.com/in/sudhir-varanasi •

PROFESSIONAL SUMMARY

Senior Software and CAE Engineer specialising in automation, simulation workflows, optimisation, and AI-enabled engineering tools for complex mechanical and multi physics systems. I combine mechanical engineering depth with practical software development to build scalable solutions across geometry preparation, HPC-scale simulation, data handling, and workflow orchestration. My recent work applies AI and data-driven methods to accelerate design optimisation, streamline analysis processes, and support predictive engineering decisions, with a growing focus on agentic AI for NX and CAE environments.

Technical Skills & Expertise:

- Artificial Intelligence, Agentic Systems & LLMOps
 - **Agentic AI & orchestration:** Architect autonomous agents using ReAct loops, tool/function calling, stateful memory, execution graph routing, MCP-based data layers, and multi-agent workflows.
 - **LLMOps, evaluation & observability:** Deploy and monitor LLM systems using LangSmith and Pydantic Logfire, with structured guardrails, automated evals, latency tracking, and reliability debugging.
 - **RAG & AI frameworks:** Build semantic retrieval pipelines using embeddings, vector databases, LangGraph, LangChain, and enterprise AI development kits.
- Scientific Machine Learning & Deep Learning
 - **SciML & optimisation:** Apply PINNs, surrogate modelling, Bayesian optimisation, multi-objective optimisation, NSGA-II, and evolutionary algorithms to engineering design exploration.
 - **Deep learning frameworks:** Develop neural networks in PyTorch and Keras; use NumPy, Pandas and SciPy for scientific computing, tensor operations, data wrangling, and matrix workflows.
 - **Advanced ML applications:** Experience with LSTMs, Autoencoders, VAEs, GANs/cGANs, time-series prediction, feature extraction, anomaly detection, dimensionality reduction, and simulation/sensor datasets.
- Software Engineering, Infrastructure & DevOps
 - **Programming & backend systems:** Python, MATLAB, VB.NET and VBScript; backend services, APIs and microservices using Django, FastAPI and Flask for engineering and AI data workflows.
 - **UI, HPC & concurrency:** PySide/PyQt desktop tools, HTML dashboards, multi-core/process parallelism, IPC, distributed workloads, memory monitoring, and HPC resource tuning.
 - **Containerisation & CI/CD:** Docker, Docker Compose, Git/GitHub Actions, automated testing, quality checks, build pipelines, and deployment workflows for engineering codebases.

- Simulation Architecture, CAE Automation & Digital Engineering Platforms
 - **CAE suites & solvers:** Ansys Mechanical, Abaqus, Simcenter 3D, Hypermesh, TOSCA, and FE-Safe for structural, thermal, dynamic, fatigue, topology and multi-physics workflows.
 - **DOE, MDO & automation:** iSIGHT and HEEDS for DOE/MDO; Python and NXOpen frameworks for parametric CAD-to-CAE integration, solver orchestration, data handling and closed-loop design exploration.

PROFESSIONAL EXPERIENCE

Rolls-Royce PLC – Derby, UK (Rolls-Royce & Infosys Strategic partnership)

Design System Automation Engineer, Jan 2025 – Present

Infosys Limited – Bangalore, India (Rolls-Royce & Infosys Strategic partnership)

Consultant, Software Methods & Data Analytics, December 2020 – Jan 2025

Rolls-Royce India Private Limited – Bangalore, India

Principal Engineer, Mechanical Methods, Dec 2015 – Dec 2020

Program Management & Strategic Initiatives:

- **Served as Onsite Coordinator and Program Manager**, acting as the primary technical interface between business stakeholders and engineering teams to identify high-value automation opportunities.
- **Managed the end-to-end commercial lifecycle** for external and internal multi-team initiatives by responding to Statements of Work (SOWs), establishing project scopes, detailing milestones, and preparing detailed delivery plans.
- **Executed procurement tasks by preparing Purchase Orders (POs)** and overseeing project execution, leading formal technical reviews to validate deliverable compliance prior to final project sign-off.
- **Served as a foundational key contributor** to the inception and rapid scaling of the Software Methods & Data Analytics (SMDA) team.
- **Led global technical recruitment** efforts by defining role-specific engineering competency frameworks, designing structured evaluation rubrics, and conducting in-depth technical interviews to assess candidates across software, automation, and CAE domains.
- **Developed onboarding and technical training frameworks** to mentor engineers in Python architecture, simulation automation, and engineering data science practices.
- **Strengthened collaboration between mechanical design and software teams** to align technical delivery with wider business and engineering objectives.

Technical Hands-On Experience:

- **Design and Development of workflow system:**
 - **Architected and deployed** an enterprise-scale Thermo-Structural Clearance Workflow (TSCW), unifying fragmented, multi-platform analysis routines into a single, coherent Python-based automation framework used for turbine seal, blade-tip, and centrifugal compressor clearance evaluations.

- **Extended the platform's core multi-physics pipeline** to evaluate seal stability margins, surge margins, and secondary flow efficiencies under transient operating cycles.
 - **Engineered coupled thermo-mechanical analysis pipelines** integrating Siemens NX (NXOpen), SC03, Simcenter 3D, Abaqus and Ansys to fully automate geometry updates, transient component temperature evaluations, and high-fidelity localized stress/strain tracking.
 - **Engineered enterprise workflow automation in HEEDS** (via Python) and **iSIGHT** (via Jython) to drive the Thermo-Structural Clearance Workflow (TSCW) execution pipeline. Built high-accuracy surrogate models for rapid design exploration and established automated post-processing routines for efficient result analysis.
 - **Introduced process-level parallelism** and intelligent resource scheduling, allowing multiple multidisciplinary simulation variants to run concurrently and accelerating overall product configuration cycles.
 - **Led the cross-platform migration** of the Windows-centric TSCW application to run natively on Linux, implementing distributed parallel-processing capabilities to take advantage of multi-node HPC environments.
- **Design Space Exploration & Optimization Architecture**
 - **Integrated advanced DOE capabilities** using Latin Hypercube Sampling (LHS) to maximize design space coverage and generate high-density training data for surrogate modeling.
 - **Constructed high-fidelity surrogate models** using Gaussian Processes, Kriging models, and physics-backed mathematical approximations to accelerate dense multi-disciplinary engineering design evaluations.
 - **Implemented evolutionary algorithms**—including NSGA-II and Particle Swarm Optimization (PSO)—to drive optimization loops via surrogate models and establish robust engineering designs.
 - **Acted as Principal Python Solution Architect** across the department, enforcing robust software engineering standards, clean code practices, and scalable architecture for computational optimisation tools.
 - **Optimized a 1 Hz high-frequency data processing tool**, yielding a **40% runtime reduction** alongside significant improvements in optimization loop latency and throughput.
 - **Full-Stack Engineering Software & UI Architecture**
 - **Designed and built a responsive PySide-based desktop GUI** to streamline end-to-end multi-physics simulation setups, dynamically generating and validating workflow configuration files to minimize user-input errors.
 - **Architected and developed a production web application** using the *Python-Django* framework to execute rigorous Six Sigma capability assessments on wax and cast manufacturing CMM data.
 - **Engineered user-friendly monitoring dashboards** using *VB.NET* and *Windows Forms* to track real-time manufacturing data pipelines and verify data-transfer efficiency from machining lines to designated archives.
 - **Engineered automated statistical data-cleaning pipelines** utilizing *NumPy* and *Pandas* to programmatically eliminate anomalies, generating interactive analytics suites (normal distributions, box/whisker plots) via *Matplotlib* and *Seaborn*.

General Motors India Technical Centre – Bangalore, India

Lead Engineer, April 2013 – December 2015

- **Sealing Design & Material Characterization**

- **Engineered and optimized** high-performance sealing systems for automobile engines, ensuring structural integrity, leak prevention, and durability under harsh thermal and mechanical operating conditions.
- **Led hyperelastic material characterization** initiatives, translating raw physical test data (e.g., uniaxial, biaxial, and planar tension/compression) into highly accurate non-linear material models (e.g., Mooney-Rivlin, Ogden) within finite element solvers.
- **Automated design and analysis** of PIP (Press in place) gaskets, **O-ring** and **RTV** gaskets for various engine applications.
- **Optimized the design of engine components** based on structural and sealing requirements.
- **Evaluated contact mechanics and sealing efficiency** under complex fastening preloads, conducting advanced non-linear structural assessments to predict seal track contact pressure and deformation.

- **Abaqus Customization & Plugin Development**

- **Developed and deployed custom Abaqus GUI plugins** and toolsets using Python and Abaqus GUI Toolkit (Abaqus/CAE) to standardize and automate repetitive modeling procedures.
- **Programmed parametric scripting routines** to automate the generation, meshing, and boundary condition setup of intricate engine seal geometries, significantly reducing preprocessing time.
- **Created post-processing automation scripts** to extract, parse, and visualize critical evaluation metrics like contact pressure profiles and stress distribution, enabling rapid design iterations.

- **Design Optimization & Tool Integration**

- **Built specialized design tools** that integrated physical engineering constraints and material limits into automated workflows, enabling robust design space exploration for sealing systems.
- **Collaborated closely with cross-functional engine design teams** to translate hardware performance issues into actionable simulation requirements and specialized software tools.

GE Aviation – Bangalore, India

Engineer, May 2011 – April 2013

- **Developed custom signal processing tools** using an Excel-based Fast Fourier Transform (FFT) framework to achieve precise modal identification for complex cyclic symmetric finite element models.
- **Executed high-fidelity elastic-plastic FEA** on engine casings using Ansys to critically ascertain ultimate bolt margins, flange capability, and transverse joint load capacities during severe Fan Blade Out (FBO) impact events.
- **Conducted cyclic symmetric modal analyses** on the PP20 engine BLISK and associated casings to establish critical operational frequencies and map accurate fan rotor-to-casing interaction margins.
- **Validated structural Campbell margins** by correlating multi-harmonic rotor and stator frequency interferences to ensure long-term structural integrity and resonance mitigation.
- **Performed cyclic symmetric frequency response simulations** across key fan and booster module components to characterize localized vibration signatures under dynamic operating conditions.
- **Engineered robust non-linear and dynamic structural workflows** to evaluate complex engine subsystem components under multi-axial, high-transient stress cycles.
- **Assessed reinforced composite structures** through comprehensive static and dynamic finite element modeling to verify compliance against aggressive weight-to-strength aerospace safety thresholds.

ISRO-SHAR – Sriharikota, India

Scientist, Sep 2009 – April 2011

- **Composite Radome Structural & Electromagnetic Design Automation**
 - **Led the structural architecture and full-scale design** of a 19-meter diameter composite radome engineered to maintain structural integrity and prevent failure under extreme cyclonic environmental loading.
 - **Engineered an end-to-end design automation pipeline** by developing MATLAB algorithms to evaluate varied geodesic configurations, automatically generating spatial point clouds to define optimal structural layouts.
 - **Streamlined CAD-to-CAE workflows** by scripting NXOpen routines that read the generated point clouds to dynamically build high-fidelity 3D models, minimizing manual geometry creation errors.
 - **Coupled aerodynamic loading with structural finite element solvers** by calculating localized pressure coefficients via CFD to estimate wind load distributions, using parametric APDL scripts to map these forces directly onto Ansys Mechanical meshes.
 - **Executed comprehensive multi-disciplinary simulations**, incorporating non-linear buckling, static finite element analysis, and modal frequency extractions to rigorously validate structural safety margins.
 - **Developed automated algorithmic routines** in MATLAB to simultaneously characterize electromagnetic loss and RF signal transmission profiles across various composite dome thicknesses and dielectric properties.

Academic Projects

IIT Kanpur

Vibration based structural health monitoring of composites:

This project focussed on characterizing the influence of delamination on the natural frequency of composite structures. Experiments were carried out on composite specimens having delamination in different locations and different ply levels. The vibration response of these structures was measured using a laser scanning Doppler vibrometer. These vibration signatures were used for finite element model updating and identifying critical damage locations.

Education

- M.Tech: Solid Mechanics & Design IIT Kanpur- 2009 — 8.75 (C.P.I)
- B.E: Mechanical Engineering, Andhra University-2007 — 82.14%